RK Health

AI Smart Patient Appointment & Medication Reminder System

Project Documentation

Submitted By

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Healthcare | Generative AI | Full Stack Web Application

HTML • CSS • JavaScript • Google Apps Script • Google Sheets • Groq AI • Twilio • Google Calendar

Internship Partner
SmartBridge SkillWallet Academy

Academic Year
2026–2027

Project Description

RK Health is an AI-powered healthcare management and patient reminder platform designed to help individuals efficiently manage appointments, medication schedules, health records, and follow-up care through a centralized and user-friendly dashboard. The application combines modern web technologies with Generative AI to simplify healthcare management and improve patient engagement.

The system enables users to schedule doctor appointments, maintain medication reminders, record personal health notes, generate AI-powered health summaries, and create printable healthcare reports. By integrating intelligent automation and cloud-based services, RK Health provides an efficient solution for organizing healthcare information in one place.

The application is developed using **HTML, CSS, and JavaScript** for the frontend, **Google Apps Script** for backend processing, and **Google Sheets** as the cloud database. It integrates **Groq's LLaMA 3.3-70B Versatile model** for AI-generated healthcare summaries, **Twilio** for SMS notifications, and **Google Calendar** for appointment scheduling. The frontend is deployed using **GitHub Pages**, while the backend is deployed as a **Google Apps Script Web App**.

RK Health offers a responsive, secure, and easy-to-use interface that helps users organize healthcare activities, receive timely medication reminders, monitor appointments, generate AI-assisted visit summaries, and maintain digital healthcare records. The project demonstrates the practical application of Artificial Intelligence, cloud computing, automation, and full-stack web development in solving real-world healthcare management problems.

Key Features

- AI-Powered Health Summary Generation
- Doctor Appointment Management
- Medication Reminder System
- Patient Health Record Management
- Google Calendar Integration
- Twilio SMS Notifications
- Cloud-Based Data Storage using Google Sheets
- Dashboard Analytics
- Printable Health Reports
- Responsive Web Interface

Scenarios

Scenario 1: Smart Appointment Scheduling

A patient logs into RK Health and schedules a doctor appointment by entering the patient's name, doctor's name, appointment date, appointment time, and clinical notes. The system validates the information, securely stores it in Google Sheets, and automatically generates a Google Calendar event link. The appointment is immediately displayed on the dashboard, enabling users to efficiently organize and manage upcoming medical visits.

Scenario 2: Medication Reminder Management

A patient adds a new medication by entering the medicine name, dosage, reminder time, frequency, and contact number. RK Health stores the medication details in the cloud database and uses Twilio SMS services to send timely medication reminders. The dashboard also tracks medication schedules, helping users improve adherence to prescribed treatments.

Scenario 3: AI-Powered Health Summary Generation

After completing a medical appointment, the patient selects the **Generate AI Summary** option. RK Health sends the appointment details to the Groq API, where the LLaMA 3.3-70B Versatile model generates a patient-friendly health summary. The summary includes the visit overview, medication guidance, follow-up recommendations, and important healthcare insights in a clear and understandable format.

Scenario 4: Comprehensive Health Report Generation

The patient accesses the AI Health Report module to review healthcare information collected within the application. RK Health combines appointments, medication schedules, health notes, and AI-generated summaries into a structured health report. Users can print or export the report for personal healthcare records or to share with medical professionals during consultations.

Technical Architecture

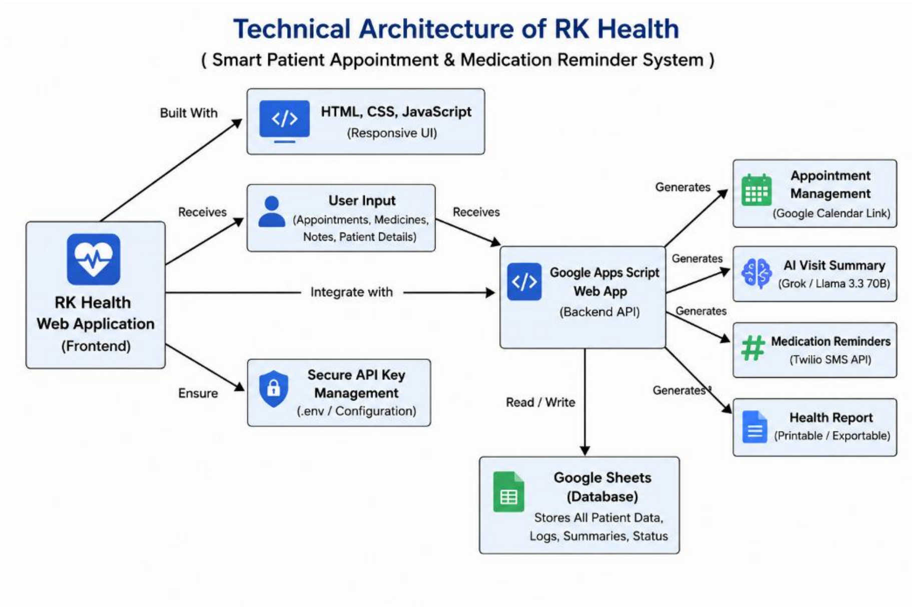

Figure4.1: Technical Architecture of RK Health – AI Smart Patient Appointment & Medication Reminder System

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Technical Architecture

RK Health follows a modern three-tier architecture that integrates a responsive web-based frontend, a cloud-based backend, Artificial Intelligence services, and external healthcare communication systems. The architecture is designed to provide secure, scalable, and efficient healthcare management while ensuring seamless communication between users and cloud services.

The frontend is developed using **HTML, CSS, and JavaScript**, providing an intuitive and responsive interface for appointment scheduling, medication management, health note recording, dashboard analytics, and AI-powered healthcare report generation. Users interact with the application through this interface to perform all healthcare-related activities.

The application communicates with a **Google Apps Script Web App**, which acts as the backend service. The backend processes user requests, validates healthcare information, manages business logic, communicates with external APIs, and performs read/write operations with the cloud database.

Patient information, appointment records, medication schedules, reminder history, health notes, and AI-generated summaries are securely stored in **Google Sheets**, which functions as the cloud database. This enables centralized data storage, real-time updates, and easy retrieval of healthcare information.

RK Health integrates the **Groq API** using the **LLaMA 3.3-70B Versatile** large language model to generate AI-powered visit summaries, medication guidance, follow-up recommendations, and patient-friendly healthcare insights. Prompt engineering techniques are used to transform structured healthcare information into concise and understandable summaries.

The system also integrates the **Twilio SMS API** to deliver medication reminders and appointment notifications. **Google Calendar** integration automatically generates calendar event links for scheduled appointments, helping patients manage healthcare activities more efficiently.

The frontend is deployed using **GitHub Pages**, while backend services are deployed through **Google Apps Script Web App Deployment**, enabling secure communication between all application components over the internet.

Architecture Components

Frontend

- HTML
- CSS
- JavaScript
- Responsive User Interface

Backend

- Google Apps Script
- API Processing
- Business Logic
- Input Validation

Cloud Database

- Google Sheets
- Appointment Records
- Medication Records
- Patient Health Notes
- AI Reports

Artificial Intelligence

- Groq API
- LLaMA 3.3-70B Versatile
- AI Health Summary Generation

External Services

- Twilio SMS API
- Google Calendar
- GitHub Pages

Pre-requisites

Before developing and deploying the RK Health – AI Smart Patient Appointment & Medication Reminder System, developers should have basic knowledge of web development, cloud services, Artificial Intelligence, and API integration. The following software, tools, and services are required to successfully build and deploy the application.

Programming Knowledge

- HTML5
- CSS3
- JavaScript (ES6)
- Google Apps Script
- REST APIs
- Prompt Engineering
- Generative Artificial Intelligence (GenAI)

Development Tools

- Visual Studio Code
- Git
- GitHub
- Google Chrome / Microsoft Edge
- Google Apps Script Editor

Cloud Services

- Google Sheets
- Google Apps Script Web App
- GitHub Pages
- Groq Cloud API
- Twilio SMS API
- Google Calendar API

AI Requirements

- Groq API Key
- LLaMA 3.3-70B Versatile Model
- Prompt Engineering Knowledge
- Secure API Key Configuration

Hardware Requirements

- Processor: Intel Core i5 (10th Generation or above) / AMD Ryzen 5 or equivalent
- RAM: Minimum 8 GB (16 GB Recommended)
- Storage: Minimum 256 GB SSD

- Stable Internet Connection (10 Mbps or above)

Software Requirements

- Windows 10 / Windows 11
- macOS Monterey or later
- Ubuntu 20.04 or later
- Google Chrome
- Mozilla Firefox
- Microsoft Edge

Project Workflow

The development of **RK Health – AI Smart Patient Appointment & Medication Reminder System** follows a structured workflow that divides the project into multiple milestones. Each milestone focuses on a specific phase of the software development lifecycle, including planning, system architecture, feature implementation, backend and frontend development, deployment, testing, and project completion. This organized approach ensures efficient development, better project management, and successful implementation of all healthcare management functionalities.

Milestone 1: Requirement Analysis and Architecture

- Activity 1.1: Generate and Configure the AI Service API Key.
- Activity 1.2: Configure Required Accounts and Services.
- Activity 1.3: Research and Select the Appropriate Generative AI Model.
- Activity 1.4: Define the Architecture of the Application.
- Activity 1.5: Set Up the Development Environment.

Milestone 2: Core Functionalities Development

- Activity 2.1: Develop Core Healthcare Management Features.
- Activity 2.2: Implement the Google Apps Script Backend.

Milestone 3: Backend and Application Development

- Activity 3.1: Writing the Main Application Logic.

Milestone 4: Frontend Development

- Activity 4.1: Designing and Developing the User Interface.
- Activity 4.2: Creating Dynamic Content Rendering with JavaScript.

Milestone 5: Integration and Deployment

- Activity 5.1: Preparing the Application for Local Deployment.
- Activity 5.2: Deploy the Frontend and Backend Services.
- Activity 5.3: Validate Deployment and Test Live Workflow.

Milestone 6: Conclusion

- Activity 6.1: Summarize the overall project objectives and achievements.
- Activity 6.2: Evaluate the performance and reliability of the healthcare management system.

- Activity 6.3: Highlight the benefits of integrating Artificial Intelligence into healthcare workflows.
- Activity 6.4: Identify project limitations and recommend future enhancements.
- Activity 6.5: Complete the final documentation, testing, and project review.

Milestone 1: Requirement Analysis and Architecture

The Requirement Analysis and Architecture phase forms the foundation of the RK Health – AI Smart Patient Appointment & Medication Reminder System. During this phase, the project requirements were carefully analyzed, the required cloud services were configured, and the overall system architecture was designed. The development environment was prepared to ensure smooth communication between the frontend, backend, cloud database, Artificial Intelligence services, and third-party integrations. Proper planning during this milestone helped establish a scalable, secure, and efficient healthcare management platform.

Activity 1.1: Generate and Configure the AI Service API Key

To enable Artificial Intelligence capabilities within RK Health, a Groq API key was generated through the Groq Developer Console. The API key was securely configured and integrated into the application to enable communication with the Groq AI service. The application uses the **LLaMA 3.3-70B Versatile** model to generate AI-powered healthcare summaries, visit overviews, medication guidance, and follow-up recommendations. The API credentials were stored securely to prevent unauthorized access while ensuring reliable communication between the application and the AI service.

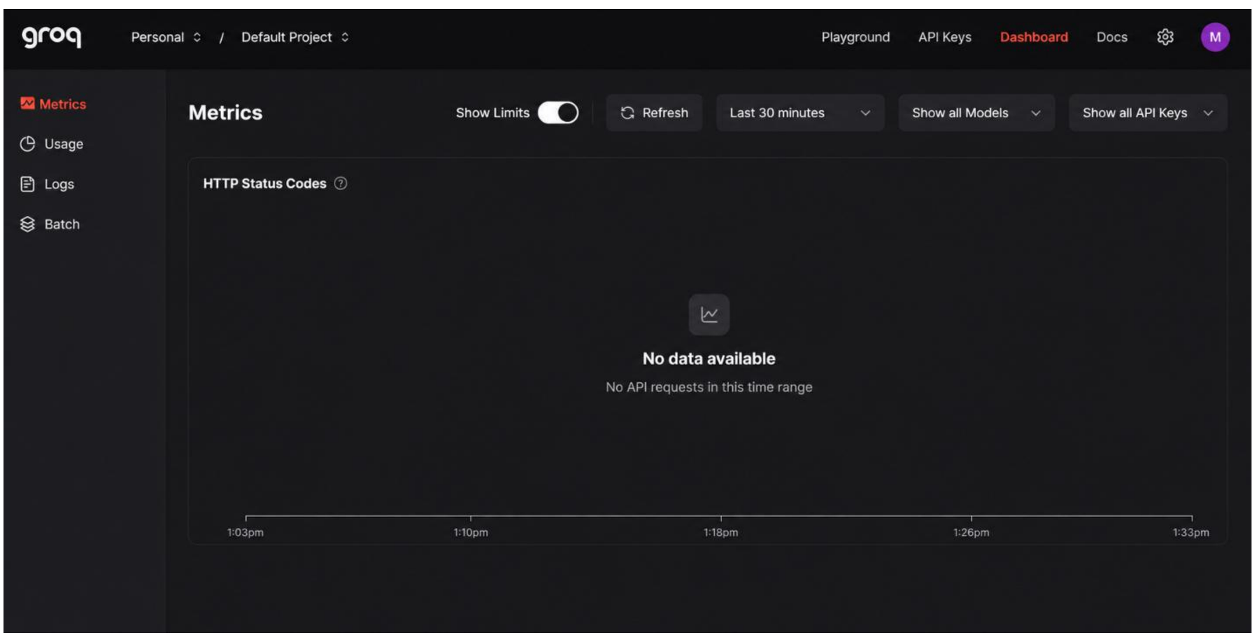

Figure 7.1: Groq Dashboard for API Key Configuration

Activity 1.2: Configure Required Accounts and Services

The required cloud services and third-party platforms were configured before the development process began. Google Apps Script was configured as the backend service, while Google Sheets was prepared as the cloud database to store appointments, medications, health notes, reminder history, and AI-generated summaries. Twilio was configured to deliver SMS notifications, and Google Calendar services were integrated to simplify appointment scheduling. GitHub was used for source code management and version control throughout the project lifecycle.

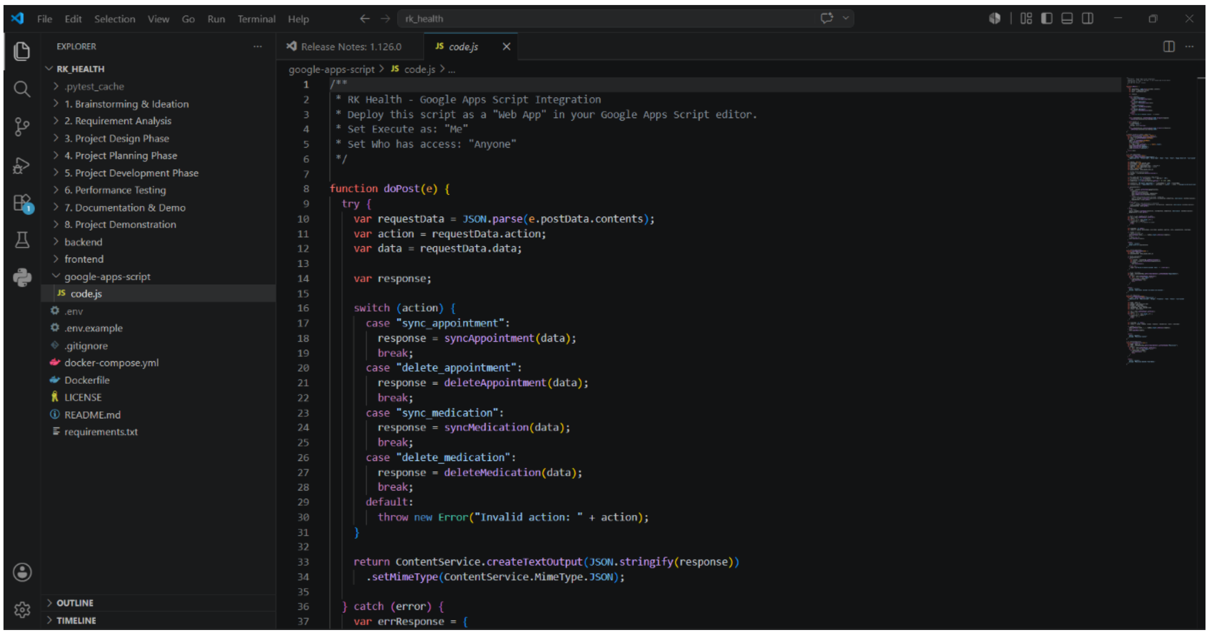

Figure 7.2: Google Apps Script Backend Project

Activity 1.3: Research and Select the Appropriate Generative AI Model

Several Generative Artificial Intelligence models were evaluated based on response quality, processing speed, reasoning capabilities, and suitability for healthcare-related text generation. After careful evaluation, the **LLaMA 3.3-70B Versatile** model provided through the **Groq API** was selected due to its high inference speed, reliable reasoning capabilities, and ability to generate clear, patient-friendly healthcare summaries. The selected model effectively converts medical notes into understandable healthcare insights for users.

Activity 1.4: Define the Architecture of the Application

The system architecture was designed to establish secure communication between the frontend, backend, cloud database, AI engine, and external healthcare services.

The frontend was developed using HTML, CSS, and JavaScript, while Google Apps Script served as the backend application layer. Google Sheets functioned as the cloud database, storing all healthcare records and application data. The architecture also integrates the Groq AI service, Twilio SMS notifications, and Google Calendar to provide intelligent healthcare management within a single platform.

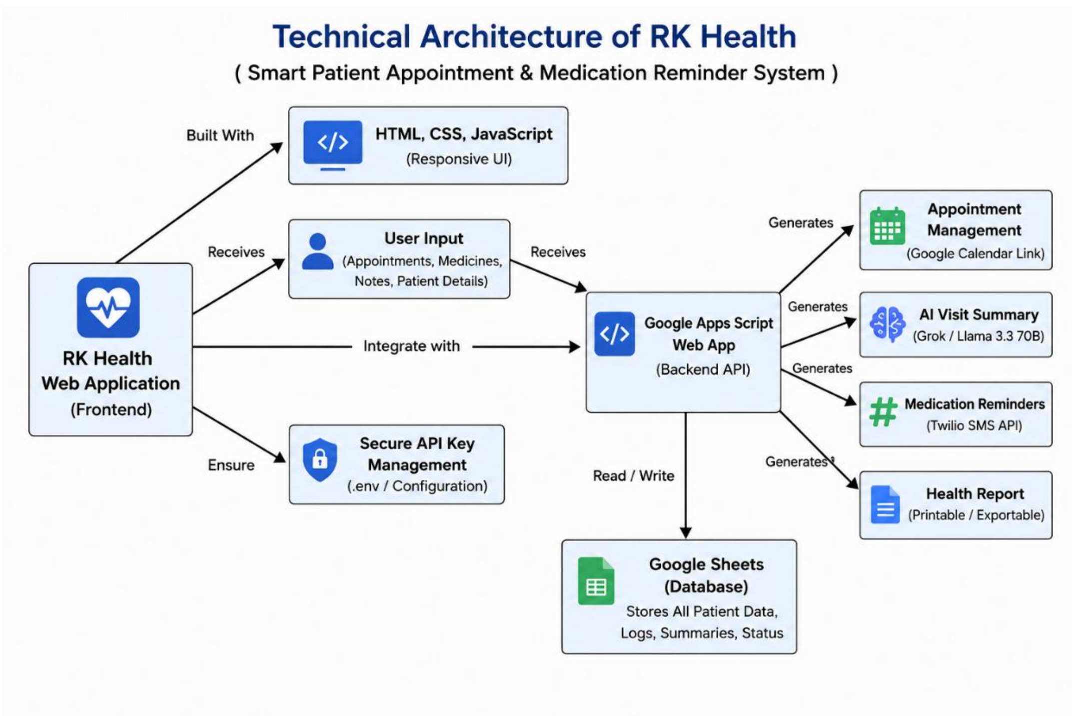

Figure 7.3: RK Health Technical Architecture

Activity 1.5: Set Up the Development Environment

The complete development environment was prepared by installing Visual Studio Code, configuring GitHub repositories, setting up Google Apps Script, connecting Google Sheets, integrating the Groq API, and configuring Twilio services. The frontend application was hosted using GitHub Pages, while backend services were deployed using Google Apps Script Web App deployment. This environment enabled efficient development, testing, debugging, and deployment of the RK Health application.

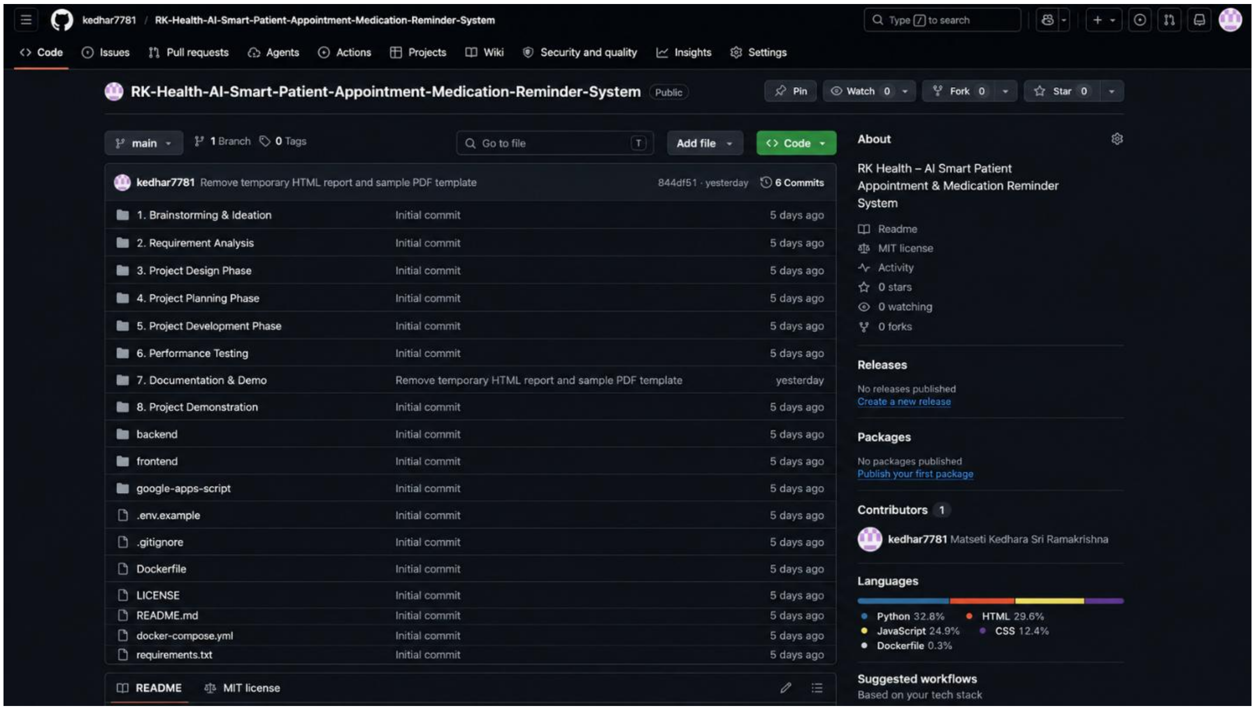

Figure 7.4: RK Health GitHub Repository

Milestone 2: Core Functionalities Development

Introduction

The Core Functionalities Development phase focuses on implementing the essential healthcare management features of the RK Health application. During this milestone, modules for appointment scheduling, medication management, health notes, dashboard analytics, and AI-powered healthcare reporting were developed. These features work together to provide users with a centralized platform for managing healthcare activities efficiently.

Activity 2.1: Develop Core Healthcare Management Features

The core healthcare modules were developed to simplify appointment management, medication tracking, health record maintenance, and AI-assisted healthcare reporting. The application provides a responsive dashboard that enables users to access all major healthcare services through a single interface. Each module is designed with a user-friendly layout to ensure smooth navigation and an improved user experience across desktop and mobile devices.

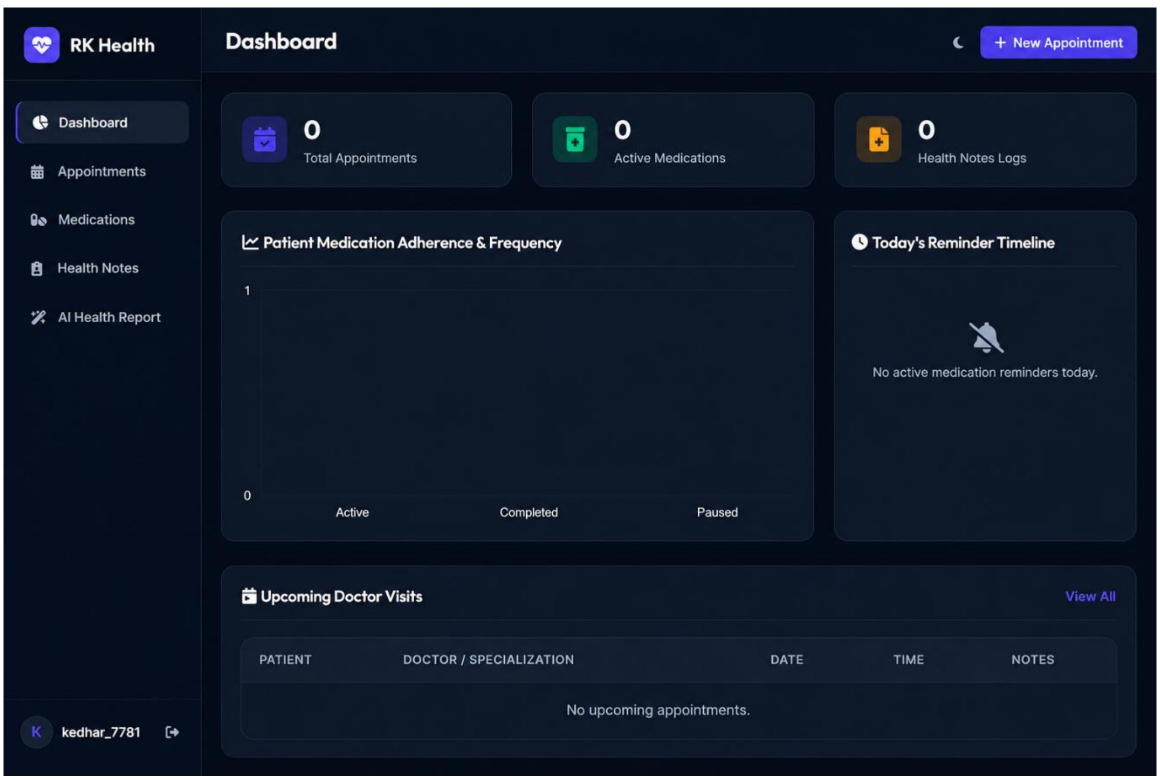

Figure 8.1: RK Health Dashboard

Appointment Management

The Appointment Management module enables users to schedule doctor appointments by entering patient details, doctor information, appointment date, time, and clinical notes. The application validates the entered information and securely stores appointment records for future reference. Integration with Google Calendar simplifies appointment scheduling and reminder management.

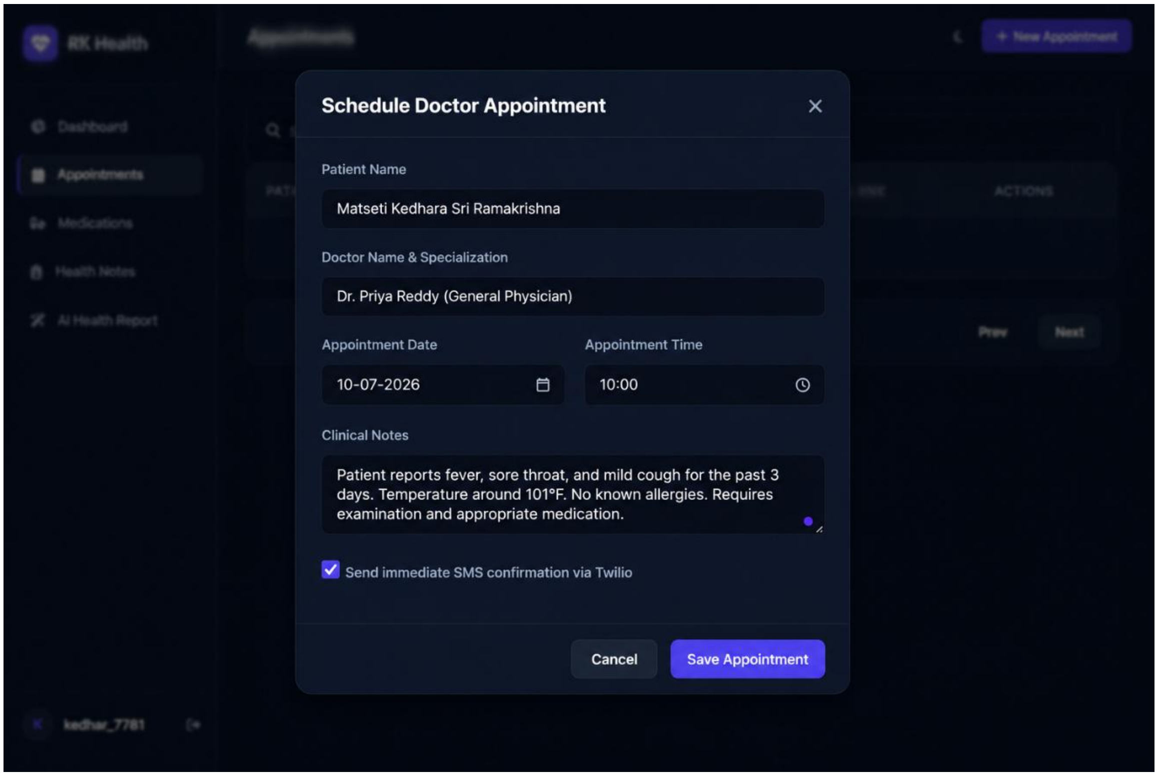

Figure 8.2: Appointment Management Module

Medication Management

The Medication Management module allows users to create personalized medication schedules by specifying medicine names, dosage information, reminder times, frequency, and treatment status. The system supports SMS reminders using Twilio to improve medication adherence and healthcare compliance.

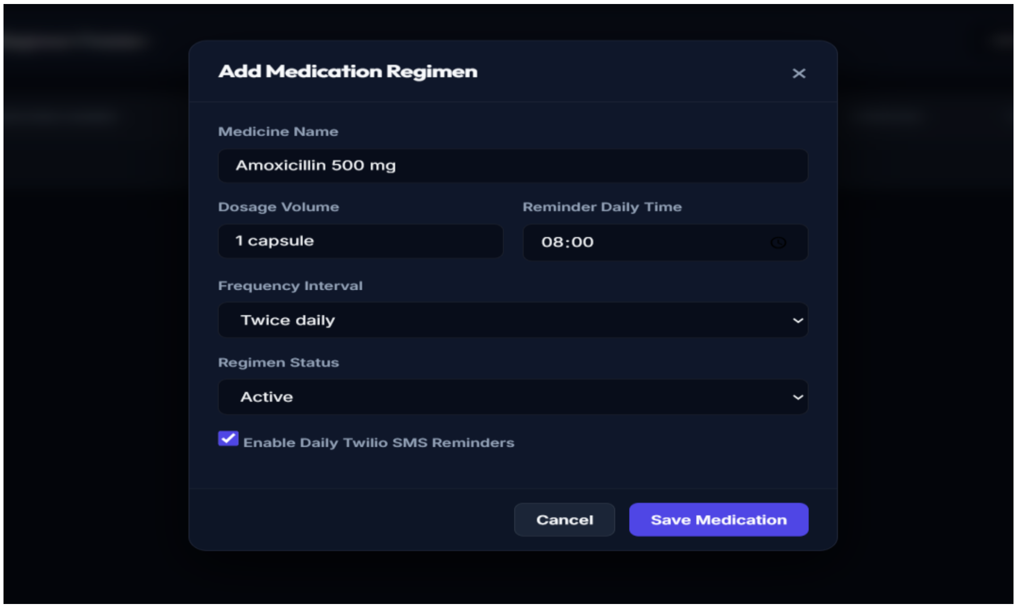

Figure 8.3: Medication Management Module

Health Notes Management

The Health Notes module enables users to maintain personal healthcare records by storing important medical observations, treatment history, symptoms, and consultation notes. This feature provides centralized access to healthcare information and supports AI-powered summary generation.

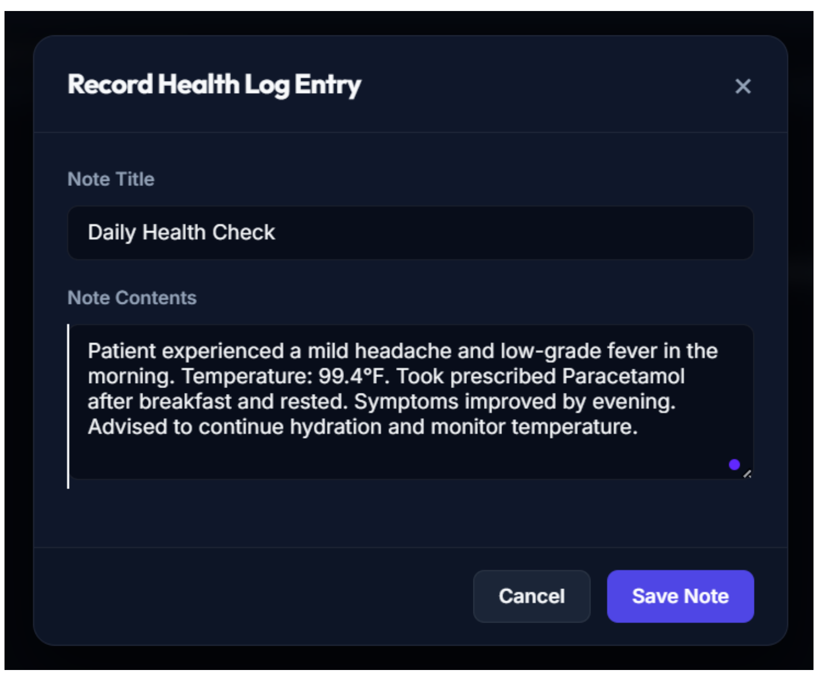

Figure 8.4: Health Notes Management

Milestone 3: Backend and Application Development

Introduction

The Backend and Application Development phase focuses on implementing the server-side logic, cloud integration, and AI-powered processing of the RK Health application. During this milestone, backend services were developed to process healthcare records, manage application workflows, communicate with cloud services, and generate AI-powered healthcare summaries. The backend ensures secure data processing, reliable storage, and seamless interaction between all system components.

Activity 3.1: Writing the Main Application Logic

The main application logic coordinates communication between the frontend, backend services, cloud database, and Artificial Intelligence engine. It processes appointment details, medication schedules, health notes, and user requests while validating inputs before storing information. The backend also integrates AI services to generate healthcare summaries and manages interactions with Google Sheets, Google Calendar, and Twilio SMS services to automate healthcare workflows.

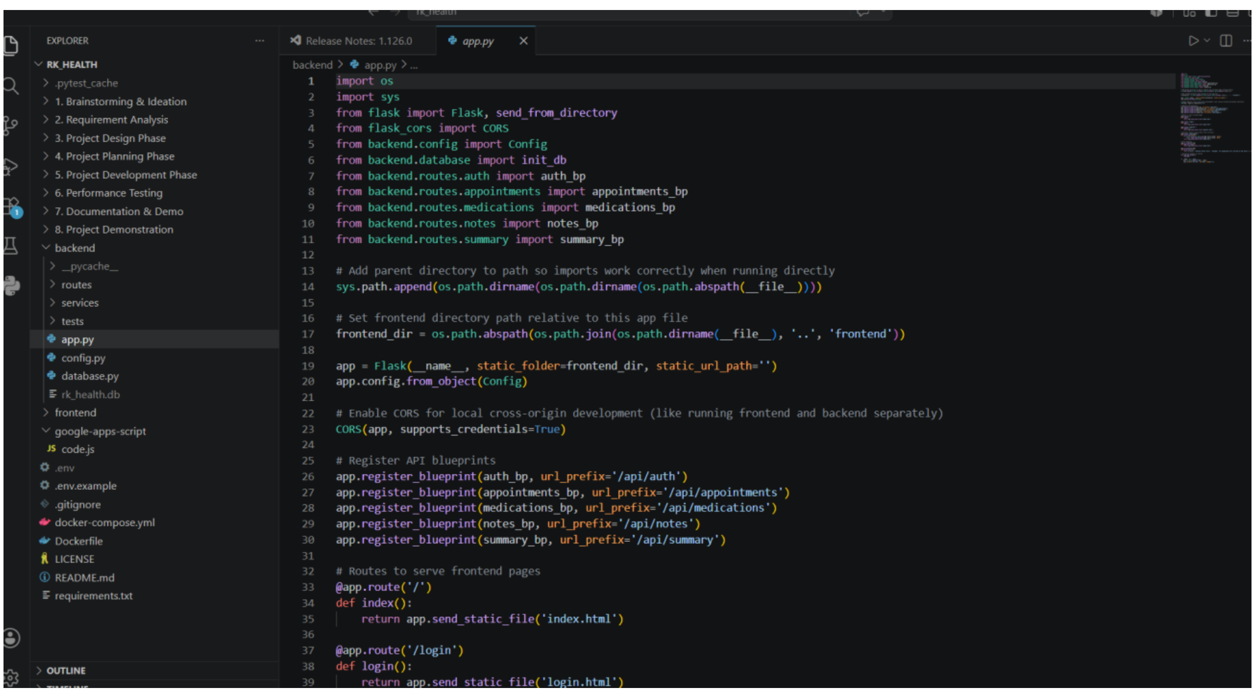

Figure 9.1: Main Backend Application Logic (app.py)

AI Health Summary Integration

RK Health integrates the Groq AI platform to generate intelligent healthcare summaries from patient appointments and health records. The backend sends structured healthcare information to the LLaMA 3.3-70B Versatile language model, which generates patient-friendly visit summaries, medication guidance, follow-up recommendations, and healthcare insights. This integration enhances healthcare accessibility by converting clinical information into easy-to-understand reports.

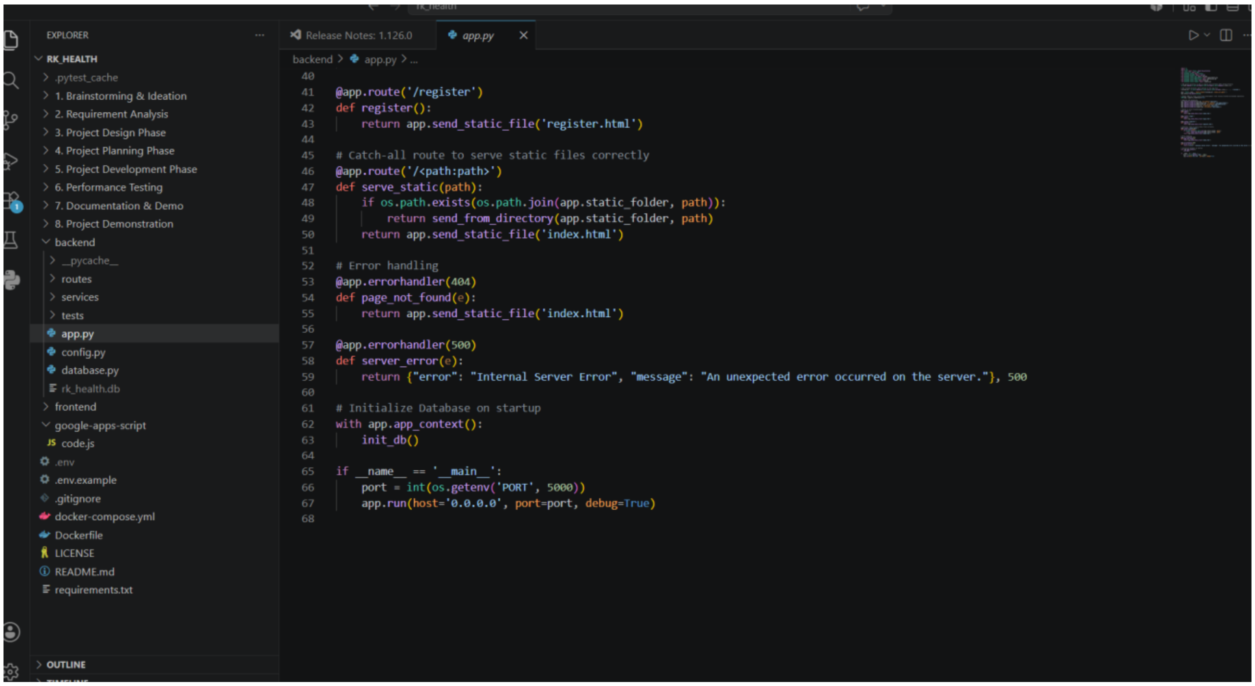

Figure 9.2: AI Health Summary Generation

Milestone 4: Frontend Development

Introduction

The Frontend Development phase focuses on building a responsive and user-friendly interface for the RK Health application. HTML, CSS, and JavaScript were used to create interactive pages that enable users to manage appointments, medications, health notes, AI-generated reports, and healthcare records. The interface was designed to provide a modern user experience across desktop, tablet, and mobile devices.

Activity 4.1: Designing and Developing the User Interface

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The user interface was designed with simplicity, responsiveness, and accessibility in mind. A modern dashboard layout, navigation sidebar, modal forms, interactive cards, and responsive components were implemented to ensure smooth navigation. Real-time updates, validation messages, and intuitive controls improve the overall healthcare management experience.

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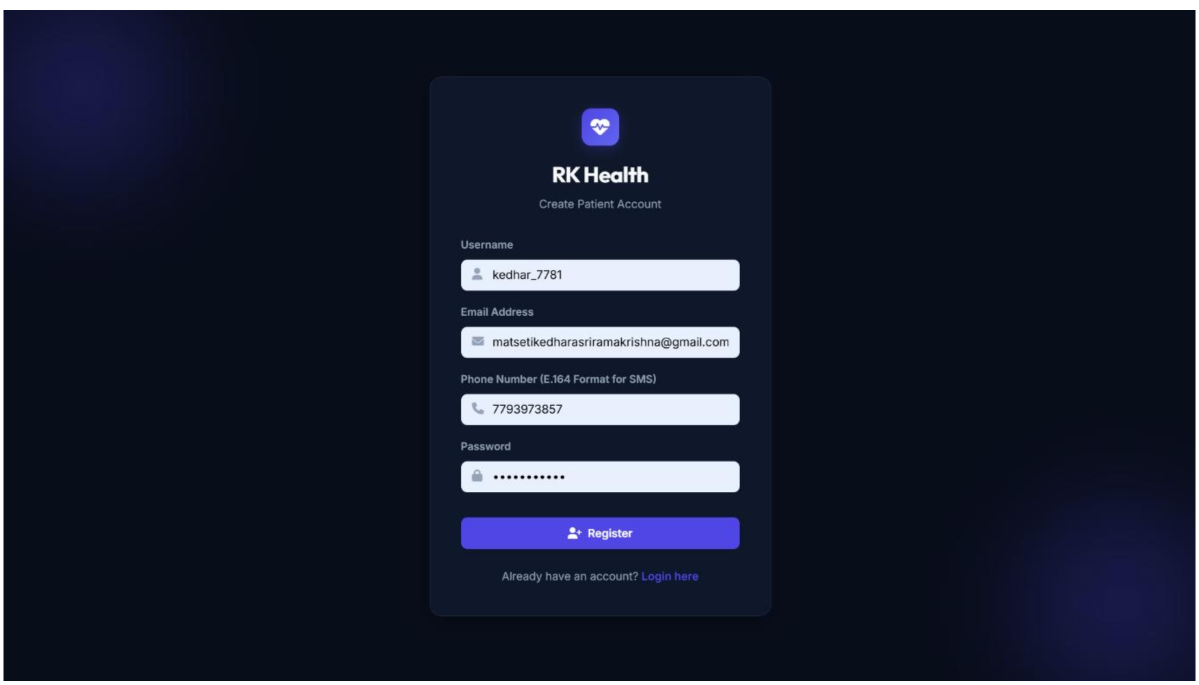

Figure 10.1: RK Health Login Interface

Dashboard Interface

The RK Health dashboard provides a centralized overview of healthcare activities, including appointment statistics, medication tracking, health notes, reminder timelines, and AI-generated reports. Interactive cards and visual components allow users to quickly monitor important healthcare information from a single screen.

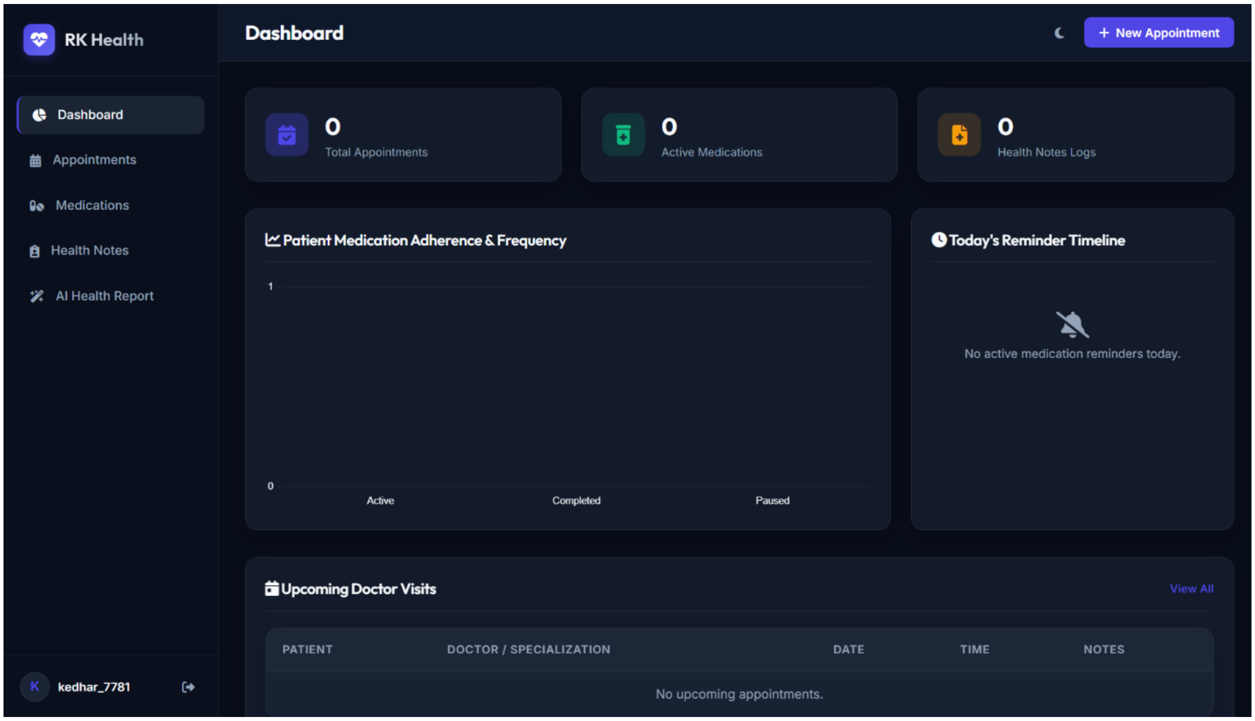

Figure 10.2: RK Health Dashboard

Responsive Design

Responsive web design techniques were implemented using CSS media queries, flexible layouts, and adaptive components. The application automatically adjusts to different screen sizes, ensuring consistent functionality and usability across desktops, tablets, and mobile devices.

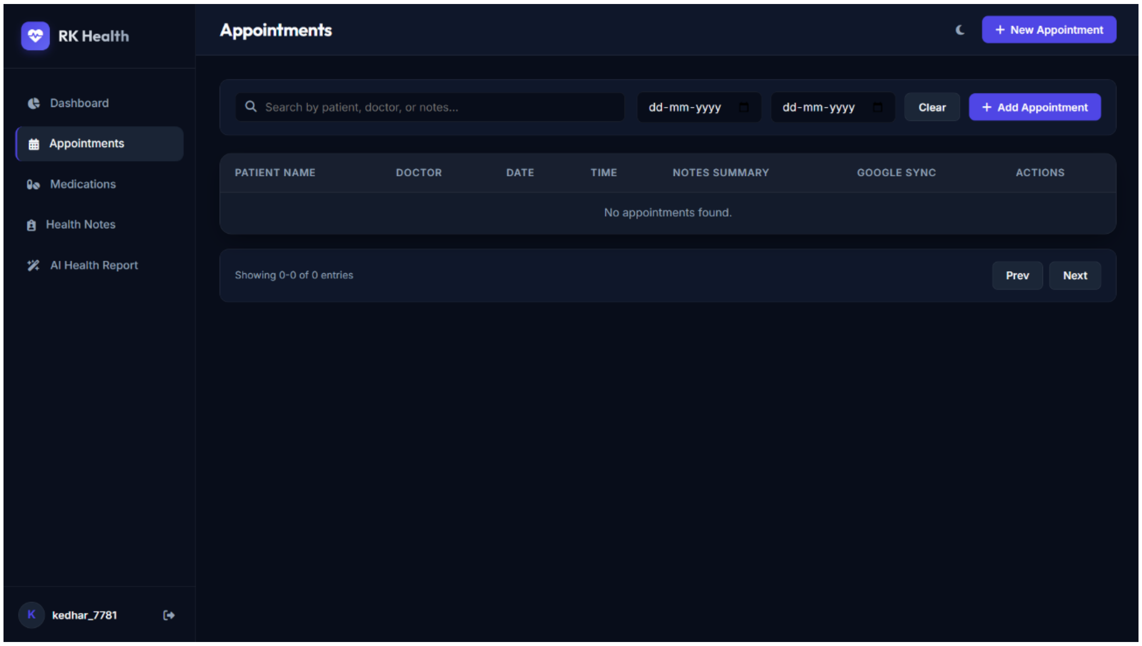

Figure 10.3: Responsive User Interface

Activity 4.2: Creating Dynamic Content Rendering with JavaScript

JavaScript was used to dynamically render appointments, medications, health notes, and AI-generated summaries without requiring page reloads. Interactive features such as modal dialogs, toast notifications, search filters, and real-time updates provide a smooth and efficient user experience.

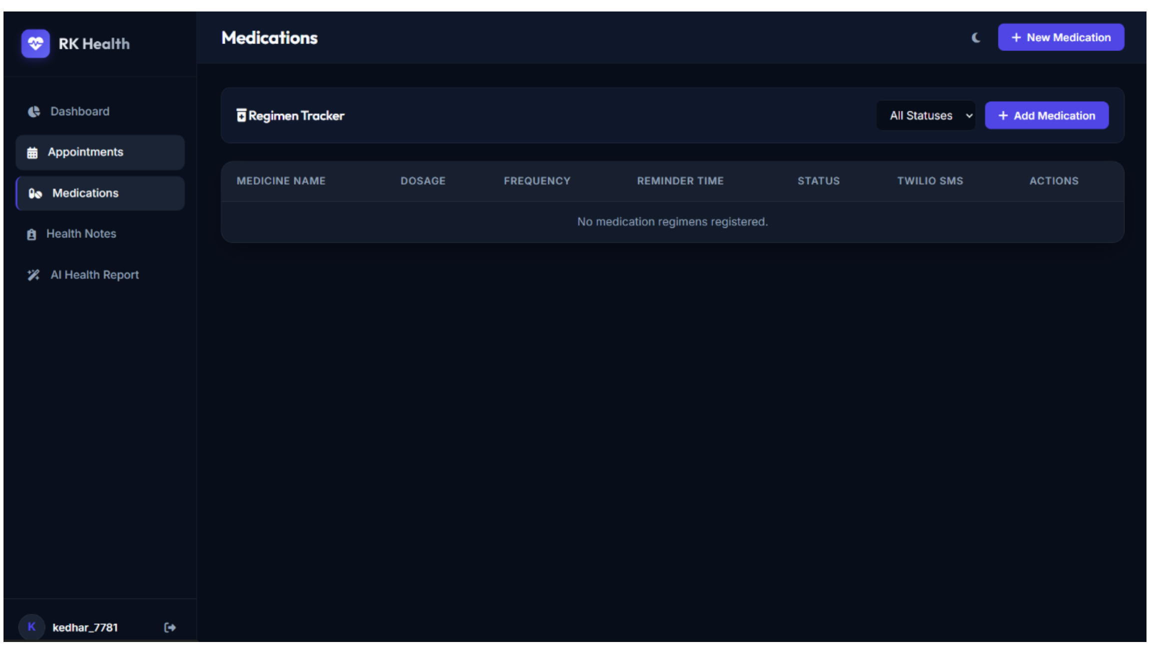

Figure 10.4: Dynamic Content Rendering

Milestone 5: Integration and Deployment

Introduction

The Integration and Deployment phase combines all frontend, backend, Artificial Intelligence, cloud database, and third-party services into a fully functional healthcare management system. During this phase, the application was deployed, tested, and validated to ensure reliable communication between all integrated services.

Activity 5.1: Preparing the Application for Local Deployment

Before deployment, the application environment was configured by organizing the frontend, backend, API configurations, database connectivity, and required dependencies. Environment variables were securely managed to protect API keys and service credentials while ensuring smooth local execution of the healthcare platform.

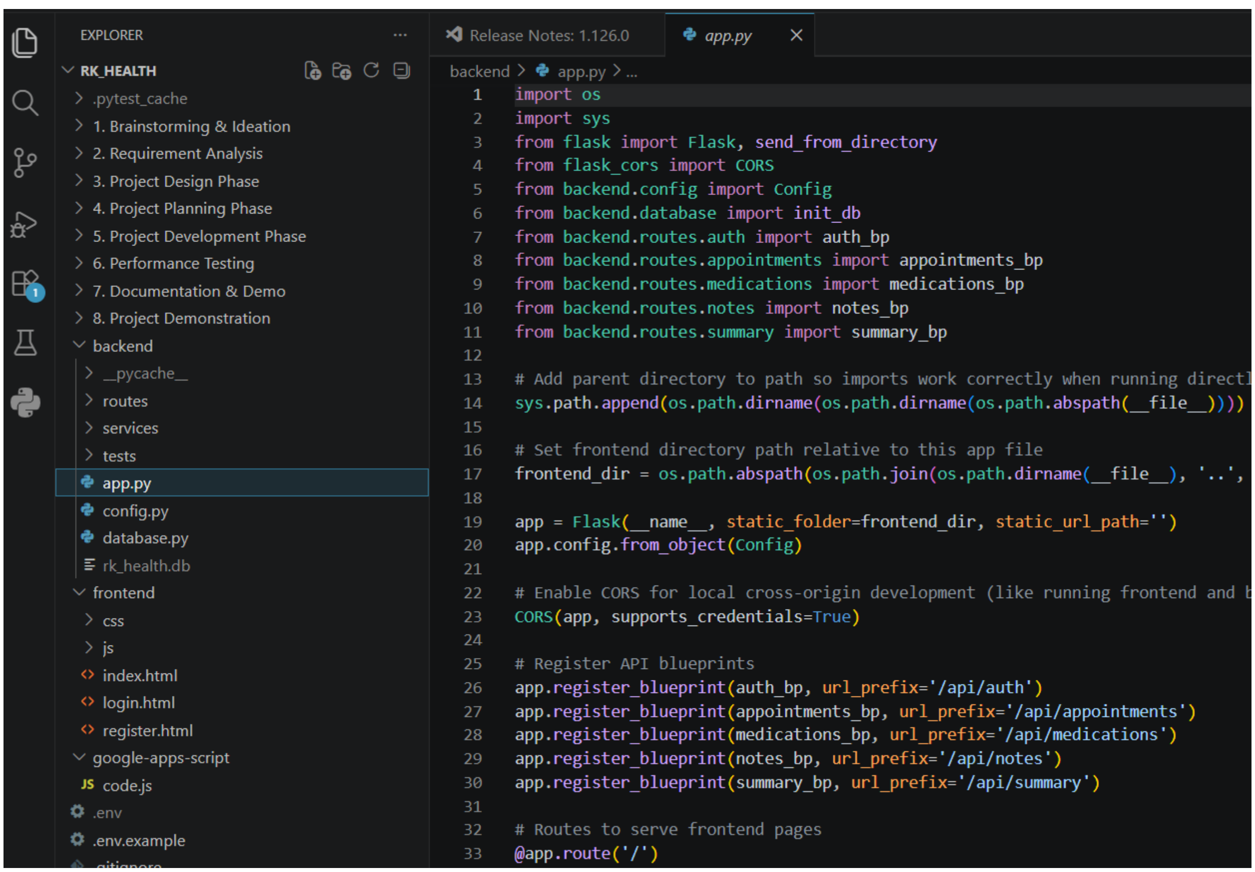

Figure 11.1: RK Health Project Structure

Activity 5.2: Deploy the Frontend and Backend Services

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The frontend of RK Health was deployed using GitHub Pages, while backend services were configured using Google Apps Script and cloud APIs. Deployment enabled secure communication between the web interface, cloud database, AI services, and external healthcare APIs for real-time healthcare management.

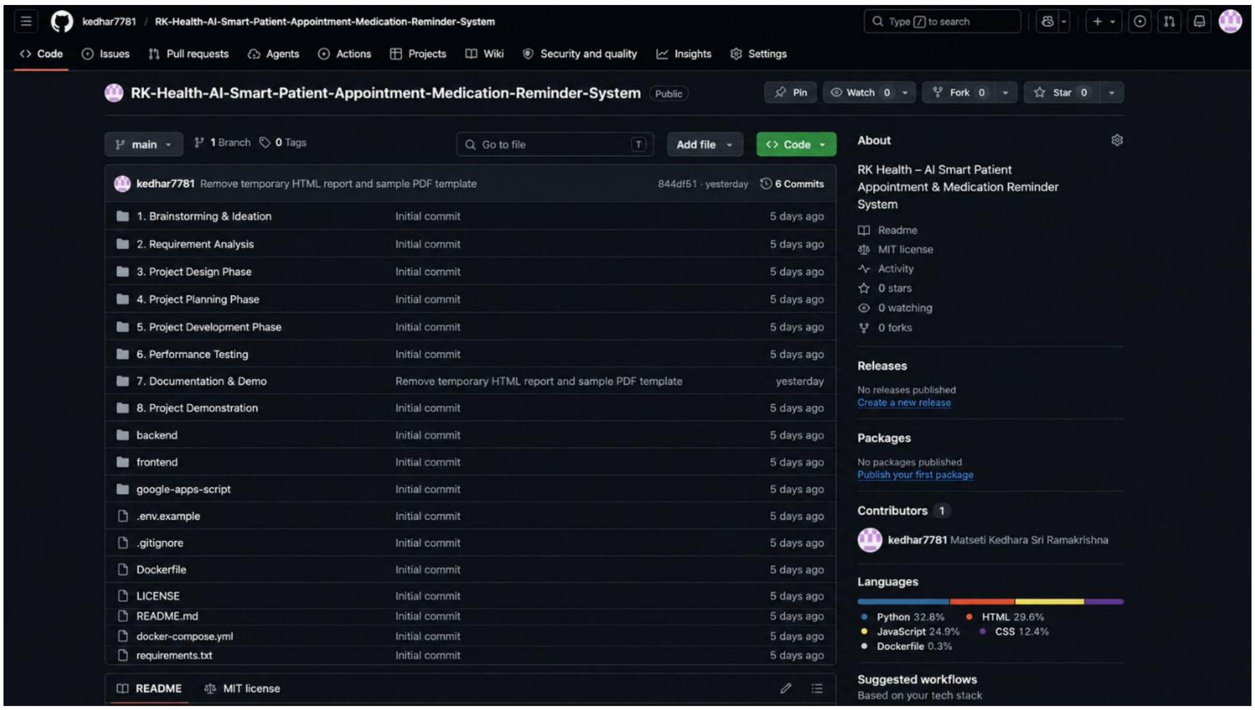

Figure 11.2: GitHub Repository Deployment

Backend Service Deployment

Google Apps Script was configured as the backend service to process appointment requests, medication schedules, AI summary generation, and cloud database operations. The backend provides secure API endpoints that connect the frontend with Google Sheets and external healthcare services.

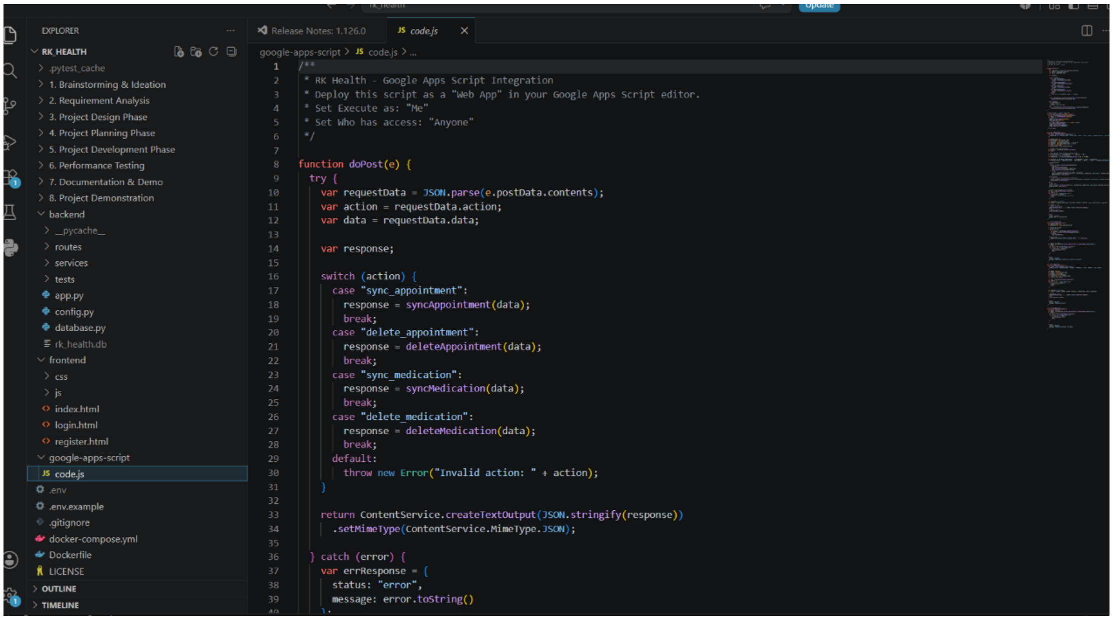

Figure 11.3: Google Apps Script Backend

Activity 5.3: Validate Deployment and Test Live Workflow

Comprehensive testing was performed to validate appointment creation, medication reminders, AI summary generation, Google Calendar integration, SMS notifications, and cloud database synchronization. The application successfully completed all functional tests, ensuring reliable healthcare management across integrated services.

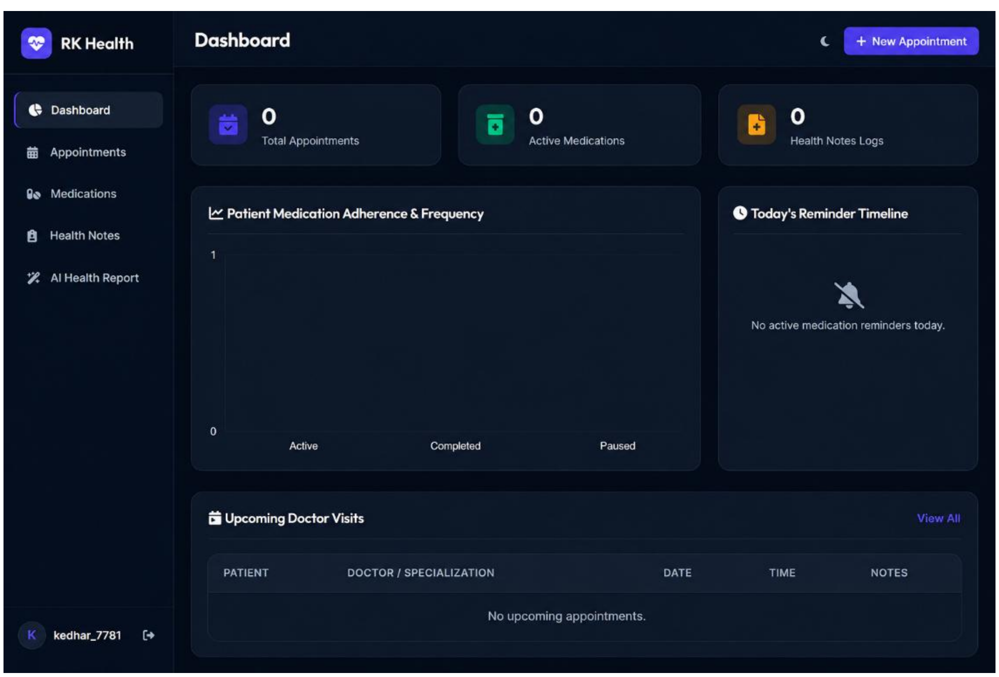

Figure 11.4: Live Application Validation

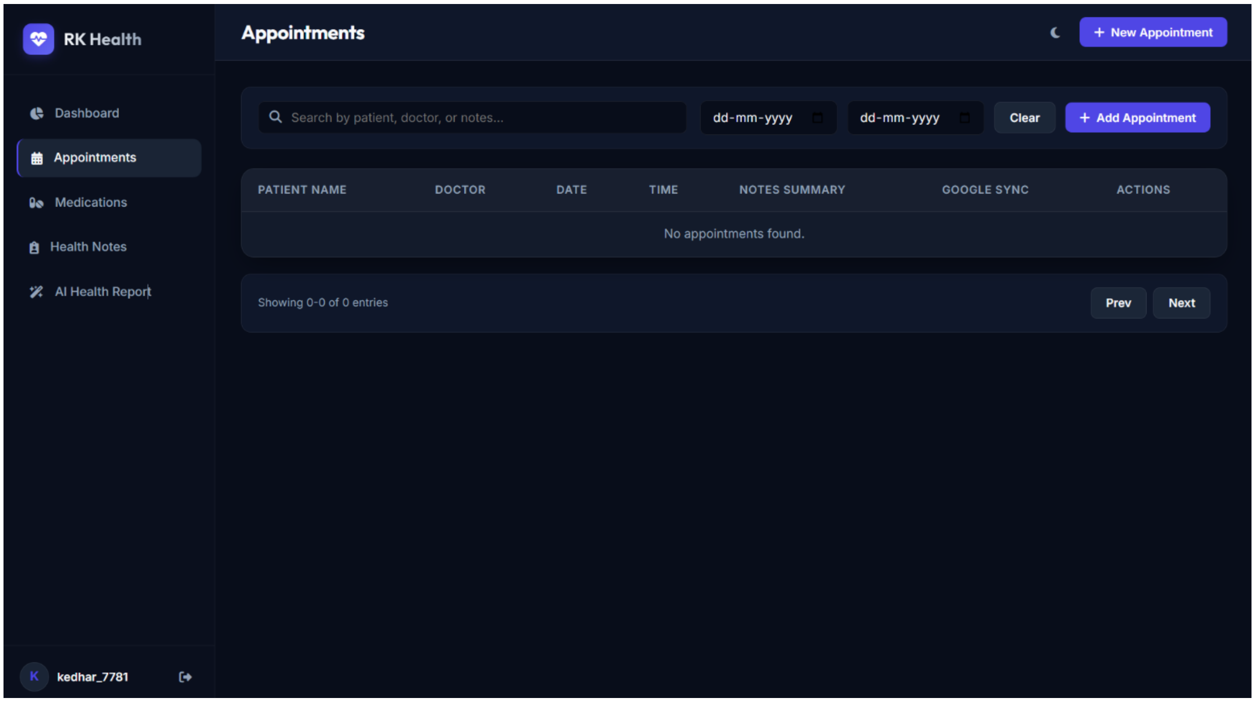

Figure 11.4: Live Application Validation

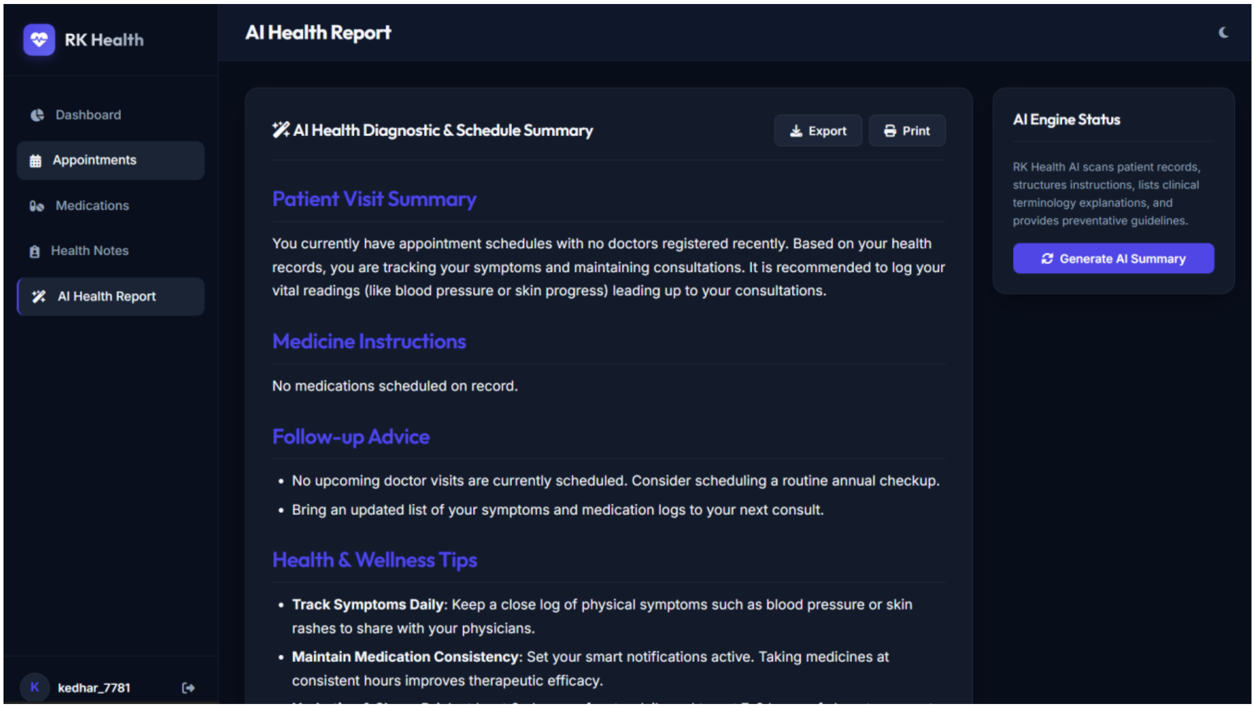

Figure 11.4: Live Application Validation

Milestone 6: Conclusion

Introduction

The Conclusion phase summarizes the successful development and implementation of the RK Health – AI Smart Patient Appointment & Medication Reminder System. The application integrates modern web technologies, cloud services, Artificial Intelligence, and healthcare management features into a single platform. The final solution provides a reliable, user-friendly, and scalable healthcare management system capable of improving appointment scheduling, medication adherence, health record management, and AI-assisted healthcare reporting.

Activity 6.1: Summarize the Overall Project Objectives and Achievements

RK Health successfully achieves its primary objective of simplifying personal healthcare management through intelligent automation. The project combines appointment scheduling, medication reminders, health notes management, AI-generated health summaries, Google Calendar integration, Twilio SMS notifications, and cloud-based services into one centralized application. Throughout the development process, the project demonstrated successful integration of frontend, backend, database, and AI technologies while maintaining an intuitive user interface and reliable system performance.

Activity 6.2: Evaluate the Performance and Reliability

The application was tested across all major functional modules, including user authentication, appointment scheduling, medication management, health notes, AI summary generation, Google Calendar synchronization, and SMS reminder delivery. Functional testing confirmed accurate data processing, responsive user interaction, secure data storage, and stable application performance. The modular architecture also improves maintainability and future scalability.

Activity 6.3: Highlight Project Benefits

RK Health provides several practical benefits, including improved appointment organization, automated medication reminders, centralized healthcare record management, AI-powered health report generation, cloud-based accessibility, and enhanced patient engagement. The application reduces manual effort, improves healthcare compliance, and demonstrates how Artificial Intelligence can enhance digital healthcare solutions.

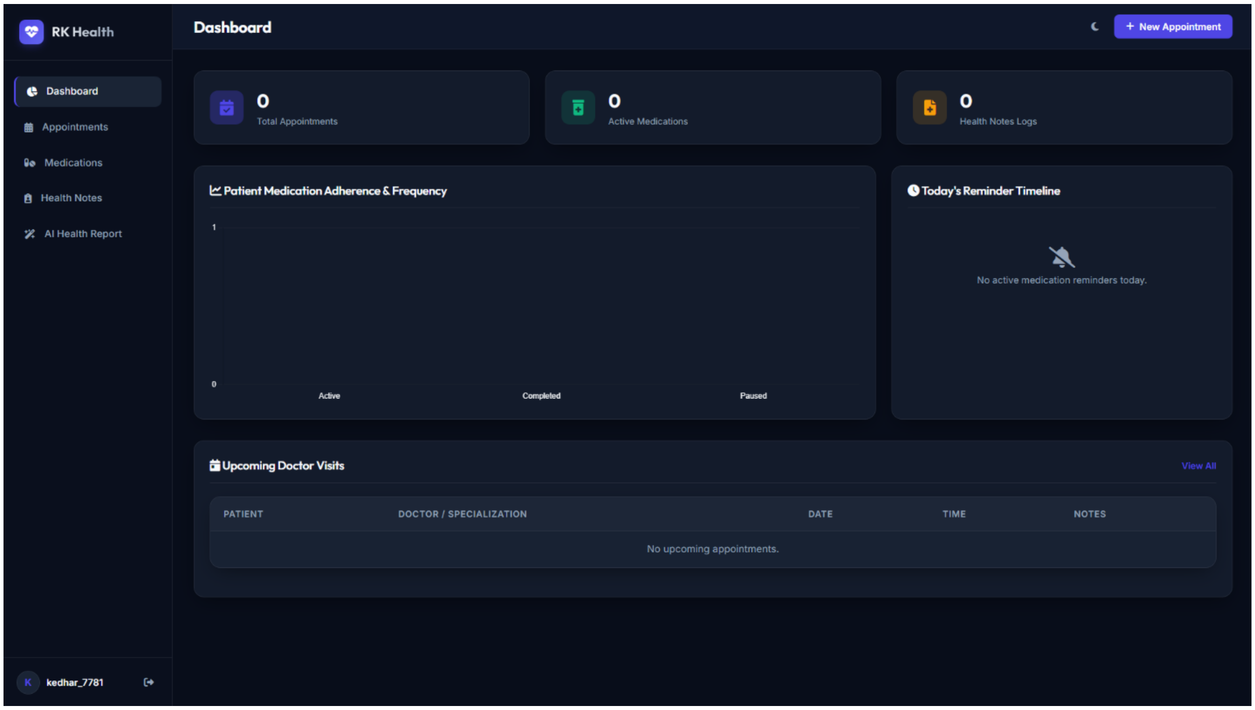

Figure 12.1: Final RK Health Application

Activity 6.4: Project Limitations and Future Enhancements

The current implementation focuses on core healthcare management features and basic AI-assisted reporting. Future enhancements may include multi-user support, doctor and hospital portals, electronic health record integration, wearable device connectivity, voice-based health assistants, multilingual support, advanced predictive analytics, and mobile application deployment for Android and iOS platforms.

Activity 6.5: Final Documentation, Testing, and Review

The RK Health project was successfully completed following a structured software development lifecycle that included requirement analysis, system design, implementation, testing, deployment, and documentation. All functional modules were verified, screenshots were documented, and the complete project workflow was reviewed to ensure accuracy, consistency, and readiness for project evaluation.